Problem No: 02
Problem Name: Echo Socket Client-Server Communication in Python

Objective:

- To implement a socket-based client-server system that echoes messages.
- To understand how the server can process and respond to client messages.

Theory:

- **Echo Server:** A server that sends back to the client exactly what it receives.
- **TCP/IP Protocol:** Ensures reliable delivery of messages.
- **Client-Server Model:**
- **Server:** Receives messages from the client and echoes them back.
- **Client:** Sends messages and displays the server's responses.
- **Port Number:** Unique identifier for the server application use port 1002 .

Key Components:

- Python socket library
- TCP protocol
- Server and client programs
- Localhost (127.0.0.1)

Application:

- Chat systems
- Network testing tools
- Message relay services

Implementation in Python:

Server Code (server1002.py)

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_socket.bind(('localhost', 1002))
server_socket.listen(1)
print("Echo server is listening on port 1002...")

conn, addr = server_socket.accept()
print(f"Connected by {addr}")

while True:
    data = conn.recv(1024)
    if not data:
        break
    conn.sendall(data)

conn.close()
```

```
server_socket.close()
```

Client Code (client1002.py)

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(('localhost', 1002))

while True:
    message = input("chithi deo : ")
    if message.lower() == 'monowar':
        break
    client_socket.sendall(message.encode())
    data = client_socket.recv(1024)
    print(f"Echo from server: {data.decode()}")

client_socket.close()
```

Result:

The client successfully connects to the server. Messages sent from the client are echoed back by the server immediately. This demonstrates the working of an echo server and reliable TCP communication between client and server.

Sample Output:

Server Console:

```
Echo server is listening on port 1002...
Connected by ('127.0.0.1', 54321)
```

Client Console:

```
chithi deo : boss
Echo from server: boss
chithi deo : ki debo
Echo from server: ki debo
chithi deo : monowar
```

Discussion :

The lab demonstrates a simple echo server where the server sends back the same messages received from the client. TCP ensures reliable communication, and the experiment shows how client-server interaction works.

Conclusion :

This lab reinforces understanding of socket programming and echo communication. Messages are transmitted and received correctly, proving successful implementation of an echo client-server system in Python.